

GIS and Machine Learning-Based Agricultural Suitability Mapping System for Konya and Surroundings

CME434 Geographic Information Systems Final Project

Konya and Surroundings, Turkiye

Alparslan Aktaş 2110213073
Osman Berke Ülger 2110213028
Fatis Selim Zeytin 2110213019

The project repo is at: <https://github.com/FatihSelimZeytin/alpberkefatihgidodev>

The web site for the project is at: <https://alpberkefatihgidodev.onrender.com/>

1. Introduction

Agricultural suitability mapping is an important spatial decision-support task for identifying areas that are more favorable for crop production. Suitability is affected by vegetation condition, topography, climate, surface moisture, and land surface temperature. In this project, a GIS and machine learning-based agricultural suitability mapping system was developed for Konya and its surroundings in Turkiye.

The project combines Google Earth Engine, satellite-derived spatial features, machine learning classifiers, and a web-based GIS application. The final system allows users to interact with a map, inspect agricultural suitability probabilities, view the spatial prediction grid, and understand the main reasons behind each prediction.

2. Study Area

The study area was selected as Konya and its surroundings in Central Anatolia, Turkiye. This region was chosen because it contains important agricultural lands as well as diverse elevation, vegetation, climate, and terrain conditions. The study area was represented using the following rectangular geographic boundary:

Boundary	Value
Minimum longitude	31.0
Minimum latitude	36.8
Maximum longitude	34.8
Maximum latitude	39.3

This boundary was used consistently in Google Earth Engine and in the web application to restrict predictions to the selected study area.

3. Label Preparation

The label preparation step was performed in Google Earth Engine using ESA WorldCover land cover data. Agricultural cropland areas were used as suitable locations, while non-cropland areas were used as unsuitable locations.

The labeling rule was:

Land cover condition	Label
----------------------	-------

Cropland	1, suitable
Non-cropland	0, unsuitable

Using stratified sampling, 943 suitable points and 810 unsuitable points were generated inside the study area. These points were merged into a single sample point dataset and exported as a CSV file.

The final training dataset contains 1753 manually labeled sample points.

4. Feature Preparation

Fifteen relevant spatial features were prepared for agricultural suitability modeling. These features represent spectral reflectance, vegetation condition, topography, rainfall, and land surface temperature.

No.	Feature	Source / Derivation	Relevance
1	B2_Blue	Sentinel-2	Surface brightness, soil and water distinction
2	B3_Green	Sentinel-2	Visible vegetation reflectance
3	B4_Red	Sentinel-2	Vegetation absorption and crop condition
4	B8_NIR	Sentinel-2	Strong indicator of healthy vegetation
5	B11_SWIR1	Sentinel-2	Soil and vegetation moisture
6	B12_SWIR2	Sentinel-2	Dry surface and moisture stress
7	NDVI	Derived from Sentinel-2	Vegetation density and health
8	NDWI	Derived from Sentinel-2	Surface and vegetation moisture
9	EVI	Derived from Sentinel-2	Enhanced vegetation vigor
10	SAVI	Derived from Sentinel-2	Vegetation index adjusted for soil background
11	Elevation	SRTM DEM	Altitude and terrain-related growing conditions
12	Slope	Derived from SRTM DEM	Terrain steepness and erosion/irrigation difficulty
13	Aspect	Derived from SRTM DEM	Terrain direction and sunlight exposure
14	Annual_Rainfall	CHIRPS daily rainfall	Water availability
15	LST_Celsius	MODIS LST	Land surface temperature and heat stress

Land cover was used only for label generation and was not used as a model input feature in the final model. This prevents the model from simply learning the label source directly and makes the prediction more dependent on environmental and satellite-derived variables.

5. Training Dataset Preparation

The 15 feature values were extracted at each of the 1753 labeled sample points using Google Earth Engine. The exported training dataset was saved in:

```
uploads/konya_agriculture_training_dataset_15_features.csv
```

The machine learning input and label files were also generated:

```
parameters/Inputs.txt
parameters/Label.txt
```

The input file contains the 15 feature columns, and the label file contains the corresponding binary suitability labels.

6. Model Development, Training, and Testing

Three different machine learning classifiers were generated and trained using the training dataset:

1. Logistic Regression
2. Random Forest
3. Gradient Boosting

Each classifier was trained twice:

1. A baseline version using initial parameters.
2. A tuned version using different parameters.

The dataset was split into training and testing subsets using a 75 percent / 25 percent split. Model performance was evaluated using accuracy, F1 score, confusion matrix, precision, and recall.

7. Model Evaluation and Comparison

The model comparison results are shown below.

Classifier	Training Stage	Parameters	Accuracy	F1 Score
Logistic Regression	Baseline	max_iter=1000, C=1.0	0.9522	0.9565
Logistic Regression	Tuned	max_iter=2000, C=0.5, class_weight=balanced	0.9476	0.9522
Random Forest	Baseline	n_estimators=100, max_depth=None	0.9658	0.9682
Random Forest	Tuned	n_estimators=300, max_depth=None, min_samples_split=4, class_weight=balanced	0.9704	0.9725
Gradient Boosting	Baseline	n_estimators=100, learning_rate=0.1, max_depth=3	0.9727	0.9746
Gradient Boosting	Tuned	n_estimators=200, learning_rate=0.05, max_depth=3	0.9749	0.9767

The best-performing model was Gradient Boosting with tuned parameters:

```
n_estimators=200
learning_rate=0.05
max_depth=3
```

This model achieved:

```
Accuracy: 0.9749
F1 Score: 0.9767
```

The best model was saved as:

```
models/agriculture_suitability_model.pkl
```

The complete evaluation output is stored in:

```
predictions/model_metrics.txt
```

8. Agricultural Suitability Mapping

After the best model was selected, a prediction grid was generated for the full study area. Google Earth Engine was used to extract the same 15 feature values for grid points across Konya and its surroundings. The prediction grid dataset was saved as:

uploads/konya_prediction_grid_15_features_no_landcover.csv

The trained Gradient Boosting model was then applied to 13,034 grid points. For each grid point, the model produced:

1. Binary suitability prediction
2. Suitability probability
3. Suitability class: Low, Medium, or High

The prediction results were exported as a GeoJSON layer:

static/predictions.geojson

The grid prediction summary is:

Suitability class	Probability range	Number of grid points
Low	< 0.40	4645
Medium	0.40 - 0.70	2155
High	>= 0.70	6234

Additional prediction statistics:

Statistic	Value
Total grid points	13034
Mean probability	0.5700
Minimum probability	0.0009
Maximum probability	0.9845

The full grid summary is stored in:

predictions/grid_prediction_summary.txt

9. Web-Based GIS Decision Support System

A web-based GIS application was developed using Flask and Leaflet. The application provides an interactive map interface for visualizing and querying agricultural suitability predictions.

The main application files are:

app.py
templates/index.html
static/index.js
static/style.css

The application includes the following features:

1. Interactive Leaflet map
2. Study area boundary for Konya and surroundings
3. Optional suitability prediction grid layer
4. Click-based location prediction
5. Suitability probability display
6. Low, medium, and high suitability visualization
7. Display of all 15 feature values for the selected grid point

8. Feature explanation popups
9. Agricultural interpretation explaining why a point is suitable or not suitable
10. Study-area boundary validation to prevent predictions outside the selected region

When a user clicks a point on the map, the system finds the nearest prediction grid point and displays its model-generated agricultural suitability result.

10. Explanation System

An explanation system was added to support decision-making. Instead of only showing a probability, the system interprets the selected location's feature values in agricultural terms.

For example, a location may be classified as not suitable due to:

1. High elevation
2. Moderate or low vegetation index values
3. High SWIR reflectance indicating dry surface conditions
4. Steeper terrain
5. Lower near-infrared vegetation response

For suitable areas, the system may explain the result using:

1. Strong vegetation signal
2. Favorable elevation
3. Gentle terrain
4. Moderate land surface temperature
5. Better rainfall availability

This makes the web application more useful as a decision support system because users can understand not only the prediction result, but also the environmental factors behind it.

11. Results and Discussion

The best model was the tuned Gradient Boosting classifier with an F1 score of 0.9767. This result indicates that the model can distinguish suitable and unsuitable agricultural areas with good performance using the selected 15 spatial features.

The prediction grid shows that a significant portion of the study area has high suitability probability, while other areas are classified as medium or low suitability. This spatial variation is expected because Konya and its surroundings contain different terrain, elevation, vegetation, and climate conditions.

The decision to remove land cover from the final feature list improved the scientific quality of the model. Land cover was used for generating labels, but not as a direct input feature. Therefore, the model relies more on satellite, terrain, and climate indicators instead of directly learning the label source.

The web-based GIS system successfully visualizes the prediction results and allows users to inspect individual locations. The optional prediction grid layer provides a spatial overview, while click-based prediction gives detailed information for specific locations.

12. Conclusion

This project developed a GIS and machine learning-based agricultural suitability mapping system for Konya and its surroundings. A labeled dataset was prepared using 943 suitable and 810 unsuitable sample points. Fifteen spatial features were extracted using Google Earth Engine, and three machine learning classifiers were trained and compared.

The tuned Gradient Boosting model achieved the best performance and was saved as a .pkl model. The model was applied to 13,034 grid points to create a full agricultural suitability prediction layer. Finally, a Flask and Leaflet-based web GIS application was developed to visualize the prediction grid, query specific locations, inspect feature values, and explain suitability results.

The project demonstrates how spatial data, remote sensing features, machine learning, and web GIS can be integrated into an agricultural decision support system.

13. Project Files

Important project outputs are listed below:

File	Purpose
uploads/konya_agriculture_training_dataset_15_features.csv	Training dataset with 1000 labeled sample points
parameters/Inputs.txt	Model input feature values
parameters/Label.txt	Suitability labels
parameters/train_model.py	Model training and comparison script
predictions/model_metrics.txt	Model evaluation and comparison results
models/agriculture_suitability_model.pkl	Best trained model
uploads/konya_prediction_grid_15_features.csv	Prediction grid feature dataset
parameters/predict_grid.py	Grid prediction script
static/predictions.geojson	Web map prediction layer
predictions/grid_prediction_summary.txt	Prediction grid summary
app.py	Flask backend
templates/index.html	Web interface template
static/index.js	Leaflet map and interaction logic
static/style.css	Web interface styling